FINAL RESEARCH PROJECT - PROCESS REPORT

Reproducibility, Provenance, and Validation

How the project binds code, data, configuration, HPC execution, and returned evidence into an auditable research record.

293	248 / 248	5
LOCAL TESTS PASSED	REMOTE FILES VERIFIED	SIGNED RETENTION RECEIPTS

VALIDATION OUTCOME

The cutover lineage is accepted, the 248-file release and remote preflight match, job 3550 returned complete evidence, and 293 local tests passed. These checks establish reproducibility of the execution boundary, not final scientific conclusions. Full screening remains unsubmitted.

TECHNICAL SUMMARY

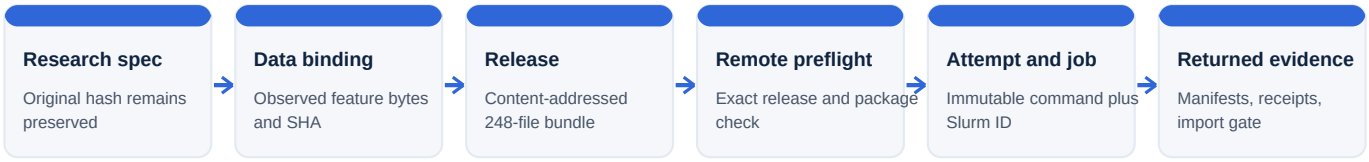
The project treats every accepted result as a chain of identities: campaign stage, immutable attempt, deployment release, remote preflight, observed feature input, resolved configuration, Slurm job, output manifest, and local import validation. The storage-safe change preserved original calibration claims through an explicit lineage receipt and rollback archive rather than silently rewriting history.

Campaign	large_scale_thai_drl_v2
Evidence cut	17 July 2026
Generated	20 July 2026
Audience	Research team, technical reviewer, and HPC operator

Evidence basis: content-addressed manifests, SHA-256 sidecars, stage success receipts, strict import validators, Slurm accounting, and the full local test suite.

SECTION 1

End-to-end provenance chain



WHAT EACH LINK PREVENTS

Binding	Evidence	Failure prevented
Campaign and stage	State and gate receipts	Using an output outside the declared dependency graph
Attempt and command	Attempt JSON and rendered-command hash	Submitting a command different from the reviewed plan
Release and preflight	Release ID, manifest SHA, returned preflight SHA	Running stale or partially copied code
Data and configuration	Feature bytes/SHA and resolved-config SHA	Comparing runs built from different inputs or parameters
Scheduler execution	Job ID, exit code, accounting, logs	Treating an incomplete or failed job as successful
Outputs	Complete manifests, retention receipts, summary and import hashes	Silent missing files, unsafe reuse, or checksum drift

LINEAGE RULE

Only control-lineage predecessor records were restored. Accepted calibration outputs, attempts, release, preflight, configuration, executable code, data, and receipt hashes remained unchanged. The lineage operation submitted zero HPC jobs.

SECTION 2

Validation layers and observed outcomes

Layer	Result	Observed evidence	Scope
Local automated suite	PASS	293 tests in 36.96 s	Code paths, release, cutover, capacity, smoke, import, and campaign gates
Deterministic release build	PASS	248 files; 7,850,129 bytes	Rebuilt twice with the same identities
Remote preflight	PASS	248 of 248 files	Release, paths, observed data, runtime packages, and feature bindings
Slurm execution	PASS	Job 3550 COMPLETED, exit 0:0	One five-algorithm storage-safe smoke
Strict import	PASS	5 manifests; 5 receipts	Checksums, paths, completeness, and policy
Checkpoint-path gate	PASS	0	No intermediate checkpoint path may survive
Campaign gate	VERIFIED	screening_smoke accepted	Advances screening to READY; does not complete screening

OBSERVED MARKET BINDING

- Feature file: data/processed/features_real_ohlcv_market.csv; 2,183,530 bytes; SHA-256 b877b08ed7a0...291ac6cc.
- Readiness: data_origin=observed_market, execution_eligible=true, evidence_class=restricted_private_research_pilot.
- The same feature SHA appears in both required runtime bindings and the returned preflight.

RELEASE IDENTITIES

Artifact or identifier	Verified value
Release ID	`56358f2a4996f654 426814d88e622b7f 217ac89bfc02f8bb dacc27e66873b55e`
Manifest SHA-256	`439b590bc4a85f03 a6d98723ee3d28e4 0acdc00d95bdfbf4 8d310f07d867e25c`
Bundle SHA-256	`a5135665dd14fdfc d0a5dcdbfab70a7e 6f88fd78fc75ff94 6df9851c01a4072e`
Preflight SHA-256	`1ccac9827e2f0189 27d711c0fb091811 9c79f8721bd6d82d 3d08ed026ab59f0c`

SECTION 3

Reproduction protocol and limitations

MINIMUM REPRODUCTION PROTOCOL

- Verify every sidecar before use; never infer a hash from a filename or a previous report.
- Require the exact release ID and manifest SHA in the remote active marker and preflight receipt.
- Verify the observed feature file by byte count and SHA before rendering an HPC command.
- Create one immutable attempt, then submit its exact rendered command. Do not edit the attempt after submission.
- Collect scheduler accounting and logs after terminal completion, fetch to a job-specific local directory, and run strict import validation.
- Promote a stage only when the declared gate passes; failed and partial evidence stays quarantined outside accepted-result globs.

WHAT IS ROBUSTLY ESTABLISHED

- The release can be reproduced and remotely verified.
- The five algorithms can execute on observed data within the smoke contract.
- The storage policy suppresses checkpoint paths while preserving final scientific artifacts and receipts.
- The returned evidence is complete and checksum-bound for job 3550.

WHAT IS NOT YET ESTABLISHED

- No algorithm is yet shown to outperform another; the smoke sample is not inferential evidence.
- No final result has yet passed full screening, aggregation, nested evaluation, robustness, ablation, and stress gates.
- Capacity evidence is time-limited. A valid historical receipt does not prove current quota or queue state.
- Reproducibility within the current software stack does not automatically establish external validity on other markets or future regimes.

AUDIT CONCLUSION

The execution boundary is reproducible and the storage-safe cutover is validated. Scientific claims remain deliberately gated until the full campaign produces accepted aggregate evidence.

PROVENANCE SOURCES

Artifact or identifier	Verified value
Accepted lineage receipt	`campaigns/large_scale_thai_drl_v2/imports/storage_safe_cutover_calibration_lineage.json`
Rollback receipt	`campaigns/large_scale_thai_drl_v2/cutovers/storage_safe_20260717_pre_cutover/snapshot_receipt.json`
Remote preflight	`reports/bistka_screening_storage_safe_cutover_v2_preflight.json`
Strict import receipt	`reports/screening_storage_safe_smoke_3550_import_validation.json`